

Predictive Analytics Assignment 3

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2026-02-13

Question 2: Role of Qualitative Predictors in Multiple Regression

```
library(ISLR)
```

```
## Warning: package 'ISLR' was built under R version 4.5.2
```

```
data(Credit)
```

(a) Regression of Balance on Gender only

```
model_a <- lm(Balance ~ Gender, data = Credit)
```

```
summary(model_a)
```

```
##
```

```
## Call:
```

```
## lm(formula = Balance ~ Gender, data = Credit)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max  
## -529.54 -455.35  -60.17   334.71 1489.20
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)  
## (Intercept)    509.80      33.13   15.389  <2e-16 ***  
## GenderFemale    19.73      46.05    0.429   0.669
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Residual standard error: 460.2 on 398 degrees of freedom
```

```
## Multiple R-squared:  0.0004611, Adjusted R-squared:  -0.00205
```

```
## F-statistic: 0.1836 on 1 and 398 DF, p-value: 0.6685
```

Interpretation:

Gender is used as the only predictor here. The coefficient for GenderFemale is positive but not statistically significant (no stars). Also, the R^2 value is almost zero (0.0005), meaning gender alone explains none of the variation in balance.

(b) Regression of Balance on Gender and Ethnicity

```
model_b <- lm(Balance ~ Gender + Ethnicity, data = Credit)
```

```
summary(model_b)
```

```
##
## Call:
## lm(formula = Balance ~ Gender + Ethnicity, data = Credit)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -540.92 -453.61  -56.37   336.24 1490.77
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      520.88      51.90  10.036  <2e-16 ***
## GenderFemale       20.04      46.18   0.434   0.665
## EthnicityAsian    -19.37      65.11  -0.298   0.766
## EthnicityCaucasian -12.65      56.74  -0.223   0.824
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 461.3 on 396 degrees of freedom
## Multiple R-squared:  0.000694, Adjusted R-squared: -0.006877
## F-statistic: 0.09167 on 3 and 396 DF, p-value: 0.9646
```

Interpretation :

This model adds ethnicity along with gender. The ethnicity coefficients are still not statistically significant. The R^2 increases only slightly to 0.001, showing that gender and ethnicity together still do not explain balance well.

(c) Regression of Balance on Gender, Ethnicity and Income

```
model_c <- lm(Balance ~ Gender + Ethnicity + Income, data = Credit)
summary(model_c)

##
## Call:
## lm(formula = Balance ~ Gender + Ethnicity + Income, data = Credit)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -794.14 -351.67  -52.02   328.02 1110.09
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    230.0291     53.8574   4.271 2.44e-05 ***
## GenderFemale     24.3396     40.9630   0.594   0.553
## EthnicityAsian    1.6372     57.7867   0.028   0.977
## EthnicityCaucasian 6.4469     50.3634   0.128   0.898
## Income           6.0542      0.5818  10.406  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 409.2 on 395 degrees of freedom
## Multiple R-squared:  0.2157, Adjusted R-squared:  0.2078
## F-statistic: 27.16 on 4 and 395 DF,  p-value: < 2.2e-16
```

Interpretation:

In this Model income is included. The coefficient of Income 6.054 is highly significant, meaning higher income is associated with higher credit card balance. The R^2 increases substantially to 0.216, indicating that income greatly improves the model.

(d) Table Output Using Stargazer

```
library(stargazer)

## Warning: package 'stargazer' was built under R version 4.5.2

##
## Please cite as:

## Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary
## Statistics Tables.

## R package version 5.2.3. https://CRAN.R-project.org/package=stargazer

stargazer(model_a, model_b, model_c,
           type = "text",
           title = "Regression Results for Credit Balance",
           dep.var.labels = "Balance")

##
## Regression Results for Credit Balance
##
=====
=====
##                                     Dependent variable:
##                                     -----
##
##                                     Balance
##                                     (1)      (2)      (3)
## -----
## GenderFemale          19.733          20.038          24.340
##                      (46.051)        (46.178)
##                      (40.963)
##
## EthnicityAsian                -19.371          1.637
##                      (57.787)        (65.107)
##
## EthnicityCaucasian           -12.653          6.447
##                      (50.363)        (56.740)
```

```
##
## Income
6.054***
##
(0.582)
##
## Constant          509.803***          520.880***
230.029***
##                  (33.128)          (51.901)
(53.857)
##
## -----
-----
## Observations          400          400          400
## R2                   0.0005          0.001          0.216
## Adjusted R2          -0.002          -0.007          0.208
## Residual Std. Error 460.230 (df = 398) 461.337 (df = 396) 409.218 (df
= 395)
## F Statistic          0.184 (df = 1; 398) 0.092 (df = 3; 396) 27.161*** (df
= 4; 395)
##
=====
=====
## Note:                                     *p<0.1; **p<0.05;
***p<0.01
```

Interpretation:

Gender and ethnicity are not significant predictors in these models, but income is a strong and significant predictor of credit card balance. Model c is the best among the three because it explains much more variation in balance.

Question 4: Impact of Ignoring Interaction Term

In this problem, we simulate data where the true model includes an interaction term: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 (x_1 x_2) +$

```
set.seed(123)

simulate_mse <- function(beta0, beta1, beta2, beta3, n=100, R=1000){

  mse_correct <- numeric(R)
  mse_naive <- numeric(R)

  for(r in 1:R){

    x1 <- rnorm(n, 0, 1)
```

```

x2 <- rbinom(n, 1, 0.3)
eps <- rnorm(n, 0, 1)

y <- beta0 + beta1*x1 + beta2*x2 + beta3*(x1*x2) + eps

model_correct <- lm(y ~ x1*x2)
model_naive <- lm(y ~ x1 + x2)

pred_correct <- predict(model_correct)
pred_naive <- predict(model_naive)

mse_correct[r] <- mean((y - pred_correct)^2)
mse_naive[r] <- mean((y - pred_naive)^2)
}

return(c(mean(mse_correct), mean(mse_naive)))
}

```

Case 1: Very Small Interaction ($\beta_3 = 0.001$)

```

result1 <- simulate_mse(-2.5, 1.2, 2.3, 0.001)
result1

## [1] 0.9631944 0.9739083

```

Interpretation:

Since the interaction term is almost zero, both models give similar MSE values. This means ignoring interaction does not cause much harm when the interaction effect is very weak.

Case 2: Strong Interaction ($\beta_3 = 3.1$)

```

result2 <- simulate_mse(-2.5, 1.2, 2.3, 3.1)
result2

## [1] 0.9577982 2.8633349

```

Interpretation

Here the interaction effect is large. The correct model has much smaller MSE. The naive model performs poorly because it misses an important relationship. Thus, when interaction terms are strong, omitting them increases prediction error significantly.